

Formation, Yield, and Persistence — Evolution by Emergence as an Optimization Principle

A working note: cost and barrier from nucleation, yield reconceived as the *reshaping of reachability*, persistence as a self-maintaining loop — with the gradient made local and non-teleological

Two refinements reorganize the back half of this note. First, **yield is released slack**: the real, budgeted, conserved capacity a structure frees when one of its loops runs more cheaply (free energy, labour-time, capital, attention — §10). Slack is a *scalar*, and that is not a problem — it is a *present, local, conserved* quantity, not a "future value." The teleology trap was never the scalar-ness of the *released* quantity; it was trying to scalarize the *downstream consequence*. That consequence — the **reshaping of which futures become reachable** — is *structural, not scalar*, and it is what the released slack *does*, not what yield *is* (§9). The corrected chain: cheaper loop → released slack (= yield) → barriers lowered → reachability reshaped → new loops possible.

Second, the mechanism is **slack-flow through loops**, and "flows toward" is formalized as *myopic diffusion of a budgeted quantity* on the edge graph (§10), removing the last suspicion of teleology by construction.

Where this note states logical entailments ("if X enters the cost term, the cube law implies Y"), those are kept crisp; hedging a deduction as one hedges an observation blurs the line between what the math entails and what we have observed, and that line is what a research program must keep sharp. *Status: the framework is **not** "everything is nucleation." It is an **optimization principle** — structures persist when what they release repays the cost of bringing and maintaining them into existence — for which classical nucleation theory is the one **rigorously solved instance** (crystals). Building the theory on the principle and exhibiting nucleation as a realization of it is far*

more robust than building it on the analogy: the principle survives even if a mapping turns out imperfect. The claim is the careful one — **many adaptive systems appear to exhibit nucleation-like barrier dynamics** — not the universal one.

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On the status of this document: it is the dense **working specification**. Its parts deliberately interlock — remove loops and enabling loses its mechanism, remove policing and the heat-death argument loses its filter, remove reachability and slack-flow has nothing to reshape — so the density is **load-bearing for its function as a reference**, not incidental bloat. It is deliberately not optimized for comfortable first reading; the cure for that is not a thinner specification (which would dissolve the interlock) but a separate, staged exposition that introduces the chain one link at a time. That companion reader-facing document exists alongside this one.

1 The primitive

The thing that governs whether a new structure — species, invention, crystal, collaboration — comes into being is a competition between two terms with a barrier between them:

Formation cost (the energy / effort / improbability of bringing

the structure into existence) versus **yield** — the capacity the structure *releases* once it exists and runs (its **released slack**, §10). The structure is worth forming iff yield exceeds cost — and between the two sits a **barrier**.

This is substrate-agnostic because "what did it cost to form this, and does what it releases repay the cost" is the *same question* for a crystal, a species, an invention, and an institution. That sameness, with content rather than vocabulary, is the claim of Evolution by Emergence.

One flag to carry forward — and it corrects a subtle error in earlier drafts. Yield (released slack) is a **scalar**, and that is *fine*: in the cleanest physical substrate (crystals, §3) it is literally bulk free energy, one number, and in every substrate it is a conserved, budgeted quantity (§10). The scalar-ness was never the problem. What cannot be a scalar is what the slack *does downstream* — the **reshaping of which futures become reachable** — and trying to compress *that* into a single "future value" number is the teleology trap §9 forbids. So keep the split clean throughout: *yield = released slack* (scalar, the cause); *reachability-reshaping* = what it pays for (structural, the effect).

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2 Two ratios, separated — and only one is the engine

- **Ratio A (Jevons / rebound — deployment)**. Efficiency rises 50%, but if usage more than doubles, total consumption rises anyway. Real and important, but a fact about *how released slack gets spent at the system level* — contingent on demand and behaviour. A downstream *consequence*, not the engine. Set aside for now (though §10 shows the slack it describes is the very thing that drives structure formation — the "consequence" turns out to be the fuel).
- **Ratio B (formation cost vs yield — the engine)**. It costs energy to make the step *exist*: 2 gallons of effort to build an improvement that saves 0.5 is a net 1.5 loss, and the improvement was a dead end. This is the cost of *forming* the pathway, not running it — the activation barrier, the "energy hurdle" from the first message. **This is the one the framework is about.**

Ratio B immediately refutes a tempting claim — "*any efficiency gain is beneficial*." No. The 2-for-0.5 case is a real efficiency gain that is **net negative**, because forming it cost more than it will ever release. Gains are beneficial **iff formation cost is repaid by yield**, and the sign can be negative. Most conceivable pathways are *not* worth forming; the rare ones that are repay the barrier, and those are what we see persist. The filter is the ratio.

3 The template: classical nucleation theory

Your ratio is **already a worked physical theory** in the cleanest substrate. A crystal forming from a melt or solution faces exactly cost-versus-yield-with-a-barrier. The free energy to form a cluster of radius r :

$$\begin{aligned}\Delta G(r) &= (\text{surface term, COST}) & + & (\text{volume term, YIELD}) \\ &= 4 \pi r^2 & - & (4/3) \pi r^3 \Delta g\end{aligned}$$

- $\gamma = \text{interfacial energy}$ — cost per unit area of the *boundary* between the new structure and its surroundings; the cost of the bonds/seams that separate "this structure" from "not it." Positive. Scales as r^2 .
- $\Delta g_v = \text{driving force}$ — free energy released per unit volume of the new phase. The (scalar) yield density. Scales as r^3 .

Because cost scales as r^2 and yield as r^3 , for *small* clusters the surface cost dominates and ΔG **rises**; for *large* clusters the volume yield dominates and ΔG **falls**. So there is a maximum — a **barrier** — at the **critical radius** r^* . Below r^* , clusters shrink; above r^* , they grow. A nascent structure below critical size *dissolves back*; only one that clears the barrier persists. The barrier height:

$$\Delta G^* = \frac{16 \pi \gamma^3}{3 \Delta g^2}$$

rises with the **cube** of the interfacial cost and falls with the **square** of the driving force. The crystal is the proof-of-concept of the whole framework: formation cost (surface) versus yield (bulk), a sharp barrier between, structure appearing **only when yield wins**.

4 The mapping across substrates — a hypothesis, not an asserted identity

The reach is the conjecture that **many adaptive systems appear to exhibit nucleation-like barrier dynamics** — the same three quantities (γ , Δg_v , ΔG^*) having *recognisable analogues* across substrates. Offered as a mapping to test case by case, not a demonstrated universality. Each row could fail; the principle survives any individual failure.

Substrate	Interfacial cost γ (boundary/bond cost)	Driving force Δg_v (yield density)	What "clearing the barrier" means
Crystal (<i>solved</i>)	surface energy of the interface	bulk free energy released by ordering	cluster exceeds critical radius, grows
Species (<i>analogy</i>)	cost of crossing the fitness valley; mutational distance to a viable adaptation	persistence the new niche-occupancy buys	a viable, reproductively coherent form establishes
Invention (<i>analogy</i>)	R&D cost; effort to assemble a working configuration	capacity the working invention releases	the thing works and propagates
Collaboration (<i>analogy</i>)	cost of forming trust-bonds between agents	durable capacity the bound collective releases	a stable cooperative holds against defection

Only the first row is established physics. The rest are *candidate* mappings, and the work (§7) is to show that for at least one, the quantities can be defined independently and predict something simpler theories do not.

5 The tweakable parameters — the three knobs nucleation theory says exist

The barrier is **not fixed**; it is a function of parameters you can move. Nucleation theory says there are exactly three knobs, and your own examples land on them.

- **Knob 1 — lower the interfacial cost γ .** Because $\Delta G^* \propto \gamma^3$, this is the *steepest* lever. Your examples — "*less hate*," "*less suspicion*," "*more honesty*" — are exactly this in the collaboration substrate: the interfacial cost between agents *is* the cost of forming trust-bonds; hate and suspicion raise γ , honesty and shared norms lower it. The cube law makes this the most powerful intervention available — reducing the cost of trust cuts the barrier *cubically*.
- **Knob 2 — raise the driving force Δg_v (supersaturation).** Push further from equilibrium — more latent, unexpressed potential — and the barrier falls as $1/\Delta g_v^2$. "*The world is soooo ready*" is a claim of high supersaturation. But high supersaturation **lowers** the barrier, it does not **abolish** it: a supersaturated solution with no seed sits metastable indefinitely. Readiness is necessary, not sufficient.
- **Knob 3 — provide a template (heterogeneous nucleation).** A foreign surface or seed **pays part of the interfacial cost for you** —

why dust nucleates raindrops, why seeding works. **Your oak proposal is heterogeneous nucleation:** bringing oak species into proximity to cross-breed both raises the supersaturation of variation (Knob 2) and provides templates — existing viable genomes a new climate-resilient configuration can assemble *onto* rather than from nothing (Knob 3).

A fact that strengthens the oak case: oaks (*Quercus*) are the textbook **syngameon** — distinct species that stay ecologically separate yet exchange genes across boundaries through hybridisation. The oak syngameon exhibits dynamics closely resembling natural heterogeneous nucleation from a shared interfertile gene pool.

6 The brakes — lowering the barrier is not free, and two examples have a known failure mode

The barrier is also a **filter**. Lower it too far and you don't get *better* structures faster — you nucleate **junk**.

- **A sweet spot for γ , not a floor at zero.** Suspicion is not *only* interfacial cost — it is also a **defector-detection mechanism**. Drop it to zero and collaborations nucleate readily, then get **eaten by free-riders**. γ cannot go to zero without removing what makes the structure durable.
 - **Too much supersaturation nucleates defective structures.** Push Δg_v too high and you get *many small unstable* nuclei rather than a few good ones. In the oak case this is **outbreeding depression**: cross too freely and you swamp locally-adapted gene complexes.
 - **The template imposes its own configuration.** A bad seed nucleates a bad structure efficiently (path dependence, lock-in).
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7 What to do next — the falsification challenge

Show, in **at least one non-physical domain**, that formation cost, driving force, barrier height, and maintenance/policing cost can be defined **independently of each other** and **independently of the outcome being predicted**, and that the framework then predicts something a simpler existing theory would not.

8 The level-jump: why stacking structure around *agents* is categorically harder

Structure-formation is **recursive**. The cost term acquires a component that exists only when the components are agents:

$$(\text{level-jump}) = \text{cost of forming the bonds} + \text{cost of POLICING the components against defection}$$

That second term **does not exist when components are inert**. If policing enters γ additively, then because $\Delta G^* \propto \gamma^3$, a high policing cost makes the barrier brutal and the transition rare.

9 What the yield *does* — the structural reshaping of reachability

Move one: Yield is released slack (§10) — the budgeted, conserved, *scalar* capacity a structure frees.

Move two: The released slack **reshapes the landscape/graph itself**, $(V, E) \rightarrow (V, E')$: the released slack lowers some barriers, and *which futures become reachable* changes.

The enabling is structural, not scalar — and that is what dissolves the teleology.

10 The mechanism — loops and slack-flow

Structure is process — maintenance is constitutive. The structure *is* the maintenance, the way a flame *is* the burning.

Slack-flow is the enabling mechanism — and it is local. Cheapening a loop **releases capacity now**, and that capacity lowers the barrier to *other* loops.

Three moments in one flow: Formation pays to *ignite* the loop. Maintenance pays to *keep it running*. Enabling is the *slack released* by running it more efficiently.

11 The network layer — two gradients that must not be merged

Gradient 1 — descent. A network moves down an **error surface**.

Gradient 2 — nucleation. Noise-assisted hop is the thermal-fluctuation barrier-crossing of nucleation.

12 The flame — what "prolonging the existence of the network" actually means

A **rock** persists *for free*. A **flame** persists *only by continuous payment* — fuel, oxygen, every instant. **A network is a flame, not a rock.**

13 What the framework is about — structure *and* flow

The framework's non-triviality lives entirely in the places where flow is *interrupted*, not where it flows. Every load-bearing element is a *resistance* to flow, a boundary, a held structure.

A structure is a loop that pays to persist *against* the flow it is made of.

14 What *kind* of claim this is — an accounting identity plus a dynamics

The framework's spine is an **accounting identity**: every persistent structure must balance a formation-and-maintenance ledger.

The falsifiable, non-trivial content is entirely in the dynamics hanging off the identity.

Stay suspicious of the warm click of recognition. The disciplines that keep this real: (1) When lowering a barrier, name which part of γ is friction and which is filter. (2) When calling something valuable for what it enables, force the claim into the reshaping-of-reachability form. (3) Never optimize reachability or viability directly. (4) Keep the gradient present-tense. (5) Lean on the shared hard mathematics. (6) Do not mistake the framework's substrate for its subject.